

Progress Report

- Increment 2 -

Group #1

1) Team Members

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2) Project Title and Description

Project Title: LibMaster

Description:

We are revamping the FSU library study room reservation system to be more accessible, easier to use, and more visually appealing. We are making it easier to visualize which rooms are available and which are not. We are adding better-querying capabilities and allowing users to sort the available rooms by library, floor, size, and time available. We will add a visualization of which rooms are available in a floor-map format.

3) Accomplishments and overall project status during this increment

In this second increment, we spent most of our time fleshing out the frontend and establishing an organized main dashboard for users of LibMaster. There were a few minor changes to our backend, which were enacted to promote a more robust organizational strategy for rooms and reservations. We established a “room-first” reservation flow for users on the frontend, so that they can place a reservation for specific study rooms across campus.

We revised the primary key identifiers for rooms. They have changed from integer UUIDs to a compound key system, which conveys information about the parent floor and library of the room at a glance. Libraries now possess a library code that is either specified by the administrator, or programmatically generated based on the library name. To accommodate the automation of library code creation, it was necessary to enforce a rule that library names must be no less than four characters. Floors also have floor codes relative to their numerical value, so standard floors (having a floor value of zero or greater) will generate room codes F0, F1, ... FN, and floors below ground level (having a floor value less than 0) will generate B1, B2, ... BK. Rooms now have a room number as well. The three aforementioned fields then come together in the Room model to create the Room’s primary key, in the form ‘LIBCODE-FLOORCODE-ROOMNUM’. This prevents any potential ambiguity or naming conflicts, and is generally more robust.

Additionally, we added a “Graduate User” user group that can be applied to LibMaster users, permitting them to view and reserve graduate-only study rooms. Consequently, Room models now support a “Graduate Only” flag, which obfuscates the room from the frontend for non-graduate users, and prevents non-graduates from accessing the reservation forms for those rooms.

We also implemented another optional restriction for Room models, `requires_administrator_approval`. If the flag is set, any student reservations that are created must be approved by the library administrators prior to the reservation status being set from “Pending” to “Confirmed”. Administrators are to do this in the devoted “Pending Reservations” page on their dedicated interface, where they can bulk-approve, deny, and delete any pending reservation requests.

Enhancements were made to the frontend authentication system. We fixed a bug that was causing an infinite token refresh loop, and ensured that authentication tokens persisted across refreshes. Generally, the authentication system is where it needs to be for the remainder of the next development cycle.

The addition of a reservation form and “My Reservation” page now allow students to manage their reservations in a comprehensive manner. They can create a new reservation for a room, or cancel a reservation that has already been created. These changes propagate to the backend database model as intended, as the reservation status is switched from “Confirmed” to “Cancelled”.

Overall, LibMaster is in a great spot for completion by the stakeholder deadline. There are only a few additional features to implement, then the application is ready for production. After we complete the interactive floor map, “time-first” reservation flow, email confirmations, library scheduling, library maximum reservation time-per-day, and analytics features, LibMaster will be complete. We are excited to bring this application’s development to a close.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

Challenge: Room Identification System

Problem: Needed a standardized way to identify rooms across different libraries

Solution: Create a comprehensive room ID system that combines library code, floor code, and room number (“e.g., DRCS-F3-307A”) for clear identification

Challenge: Graduate-Only Room Access

Problem: Some rooms are restricted to graduate students only.

Solution: Implemented permission checking based on user groups, with frontend and backend validation to prevent unauthorized reservations.

Challenge: Auth tokens not refreshing as needed on the frontend

Problem: The auth error sent by the backend is 401, not 403, the options of the request have not persisted, interceptors not catching the proper error cases.

Solution: A lot of debugging, to find all the bugs in the auth system, specifically the frontend interceptors, setting the expiration of the access token to 5s to debug refresh more efficiently.

Challenge: Implementing Unique Hours of Operation for Libraries

Problem: making a usable interface for admin to be able to easily edit hours of operation.

Solution: Used Django Administration page tools to add an opening time and closing time to a library for each day of the week, which is easy for admin to update.

5) Team Member Contribution for this increment

a) Progress Report Contributions:

- i) **Dylan:** I completed the *Plans For Next Increment* section, in which I outlined the remaining features that need implementation during the duration of the final increment. Additionally, within this section, I devised a plan that describes how we will go about implementing the interactive library floor visualizer. I also completed the *Accomplishments and Overall Project Status During This Increment* section in its entirety, detailing all completed goals, and the state of LibMaster throughout this particular portion of the development cycle.
- ii) **Ivan:** Described another challenge that we faced and contributed to the increment video.
- iii) **Zach:** I added two challenges that we faced in the current increment. I also took care of writing the stakeholder email.
- iv) **Colby:** Added a challenge to the challenge section and contributed to the video.

b) R&D Document Contributions:

- i) **Dylan:** Altered the UML use case diagram to accommodate the new use case for students requesting a room reservation, and the new use case for administrator users which enable them to parse incoming pending reservations.
- ii) **Ivan:** Revised and augmented the functional requirements for users and admins
- iii) **Zach:** I added to the usability section in the non-functional requirements.
- iv) **Colby:** Revised part 2 student functional requirements, helped with part 4 use case diagram.

c) I&T Document Contributions:

- i) **Dylan:** No contributions of note for this particular increment. My focus was primarily on the Progress Report and R&D Document, and it is my hope that these contributions, in addition to those made to the source code, will be significant enough to excuse this fact.
- ii) **Ivan:** Revised functional testing methodology for auth and tokens
- iii) **Zach:** I updated the technologies section with new libraries that were added.
- iv) **Colby:** Updated part 3 student requirements, part 4, and part 5.

d) Source Code Contributions:

- i) **Dylan:** Engineered the “room-first” reservation form, and the My Reservations page in their entirety. Implemented new room primary key identification system, using hierarchical model attributes to form a compound key. Configured the room permission attributes to function on the frontend and backend, with new ‘Grad Only’ and ‘Req.

Approval' flags to be set on a per-room basis. Set up the administrator interface to allow admins to oversee the approval or denial of pending room reservations. Ensured that undergraduate students cannot access graduate-specific study rooms by exposing a user's role to the frontend, through configuration of AuthProvider's Role enumeration.

- ii) **Ivan:** Set up Graduate Students group and room flag, debugged and fixed obscure auth token bugs (infinite refreshes, no refresh without a page reload, etc)
- iii) **Zach:** Added error visualization to the signup page for users
- iv) **Colby:** Implemented library schedules, allowing admin to set hours of operation for each library. I also updated the reservation system to accommodate this change, displaying the appropriate error message to a user who tried to book a room outside of library operation hours.

6) Plans for the next increment

Given that the LibMaster development cycle is nearing its final increment, we plan to achieve a full implementation of the remaining desired functionality for our application. This includes several key features, such as the interactive library floor map, an additional time-first reservation flow to complement the existing room-first reservation flow, email-based authentication for new LibMaster users, comprehensive library scheduling across the entire year with support for holidays, and general UX improvements across the application. We also want to implement email confirmations for successfully confirmed room bookings.

At a high level, the interactive floor map visualizer should allow the user to specify a non-overlapping time interval to visualize a floor. It should then mark rooms as red, yellow, or green, corresponding to fully unavailable during the specified interval, partially unavailable during the specified interval, and fully available during the specified interval, respectively. Hovering the mouse over the room should (a) list any conflicts if they exist, and (b) if a partial conflict exists, specify a similar interval that is available (same length, in an interval no greater than +/- 30 minutes from original specification), and (c) if no conflicts exist, submit a "room-first" reservation request for that room during the interval specified in the interactive floor map.

Given the rather rich set of features we want the interactive library floor map to offer, we have assembled a list of dependencies that we will utilize to implement them, in addition to the native scalable vector graphics support offered by React: `'date-fns'` as a lightweight date utility library for time interval comparisons, `'@tippyjs/react'` for advanced tooltips when showing conflicts and alternatives, `'react-svg-pan-zoom'` for navigation of large floor plans, and `'framer-motion'` for smooth hover effects and state transitions.

7) Stakeholder Communication

Dear Stakeholders,

I hope you are doing well. I am excited to share our development team's progress on LibMaster since our first implementation increment. We have expanded both the front-end and back-end capabilities significantly while resolving several challenges that emerged during this development phase.

We have implemented several critical new features, including a complete reservation system that allows users to book rooms for specific time slots, a "My Reservations" page for users to manage their existing bookings, room amenity indicators (whiteboards, monitors, windows), and graduate-only room restrictions. The system now supports special approval workflows for rooms requiring administrator confirmation and includes a Recent Reservations component on the main page for quick access to current bookings.

Some challenges we encountered and overcame included: authentication persistence across page refreshes, which we solved by implementing a token refresh mechanism that automatically renews JWT tokens; complex reservation validation to prevent double-bookings, which required carefully designed backend logic with multiple validation layers; and handling rooms with special restrictions (graduate-only, administrator approval), which we addressed through both frontend and backend permission checks.

In our next round of development, we plan to implement a visual interface for room availability, enhancing the user experience when selecting time slots and reserving specific locations. We also plan to add email notifications for reservation status changes and develop an analytics dashboard for administrators to monitor room usage patterns and identify optimization opportunities.

We appreciate your continued support and insights as we move forward. Please let us know if you have any specific questions or input at this stage. We look forward to sharing more progress in our next update.

Best regards,

The LibMaster Development Team (Zack Lima)

8) Link to video

https://youtu.be/vsgg2Z_X5Ds